

Logic Gates 1847 UNIT - 2

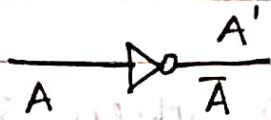
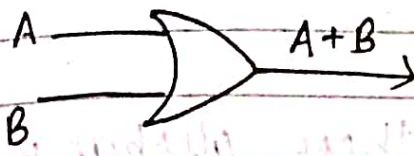
Logic gates are the fundamental or a basic component of digital electronics which is used to perform logical operations.

A logic gate may have one or more inputs but it always generate only one output.

Types of logic gate.

Primary

- i) OR (+)
- ii) AND ($\cdot$)
- iii) NOT ($-$, $'$)
(1i/0)



Truth Table

2 i/p OR

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

3 i/p OR

A	B	C	A + B + C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

properties

- When any one input is 1 then output will be 1.
- When all the input is 0 then output will be 0.

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

Properties.

- When all inputs are 1 then output will be 1
- When any one input is 0 then output will be 0.

NOT

A	A'
0	1
1	0

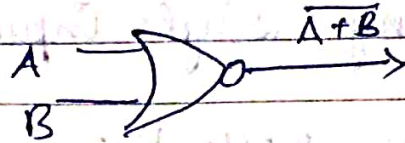
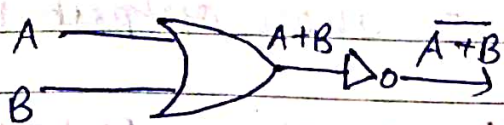
Secondary

NOR (OR + NOT)

NAND (NOT + AND)

XOR (exclusive - Ex - OR)

X-NOR



A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

Properties

- If any single Input is 1 then output is 0.
- When ~~at~~ all the input is 0 then output is 1.

OR - 4071 / 7432

NOR - 7402

NAND - 7400

NOT - 7404

XOR - 7486

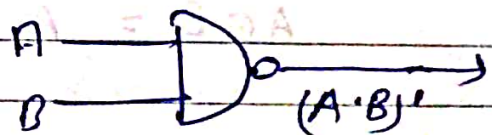
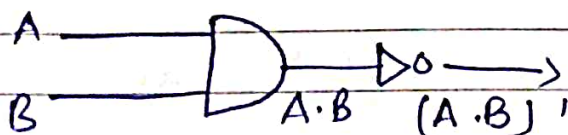
AND - 4081

70 7408

CMOS / TTL.

↳ Complementary metal
oxide Semiconductor

Transistor - Transistor
logic.

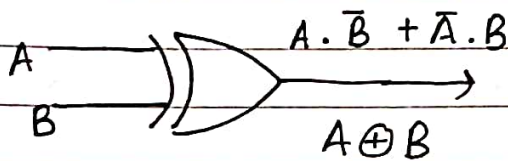


A	B	$(A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

Properties

- If any single input is 0 then output is 1.
- When both the input is 1 then output is 0.

X-OR



	A	B	$A \oplus B$
	0	0	0
odd	1	0	1
1	0	1	1
2	1	1	0

$$A \cdot \bar{B} + \bar{A} \cdot B$$

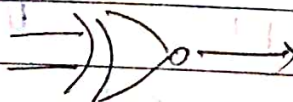
$$0 \cdot \bar{1} + \bar{0} \cdot 1$$

$$0 \cdot 0 + 1 \cdot 1$$

$$0 + 1$$

$$1$$

$$A \odot B = A \cdot B + \bar{A} \cdot \bar{B}$$

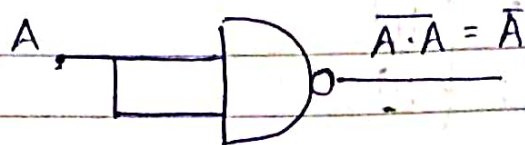


	A	B	$A \odot B$
even	0	0	1
odd	0	1	0
odd	1	0	0
even	1	1	1

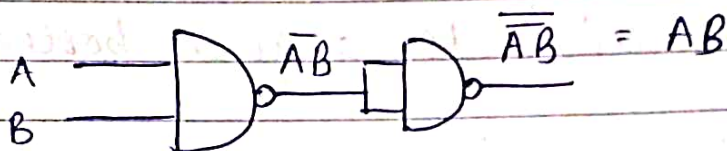
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NAND as Universal logic gate.

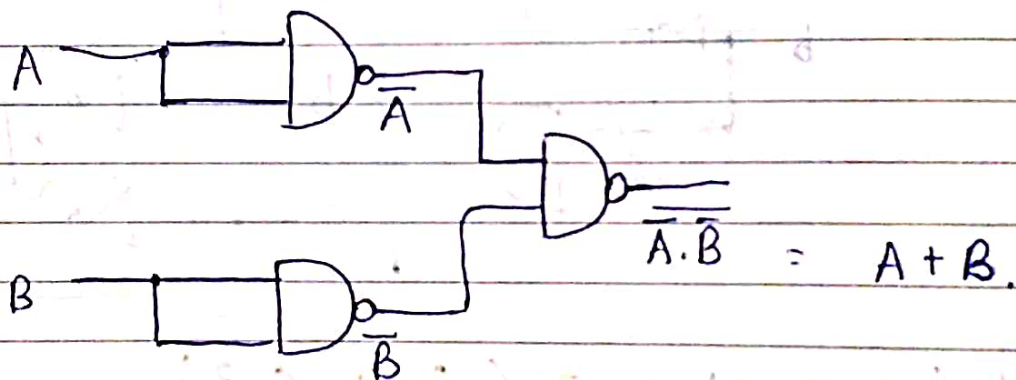
NOT Gate using NAND



AND Gate using NAND



OR Gate using NAND



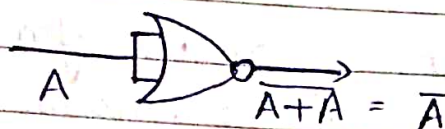
UNIVERSAL LOGIC GATES.

• NAND and NOR are universal logic gates.

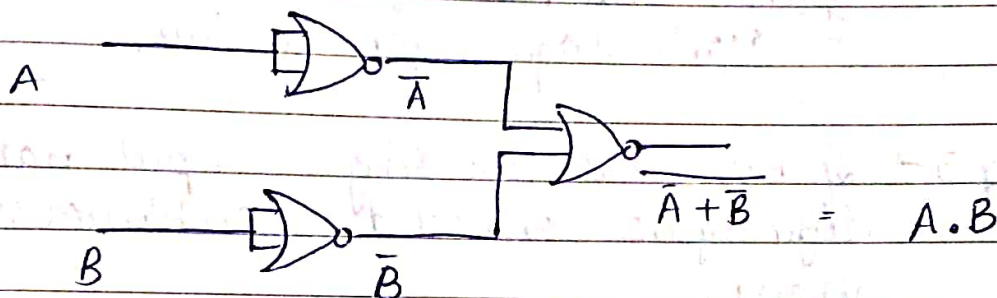
• Because only by using NAND gate all primary logic gates can be implemented.

NOR as universal logic gate.

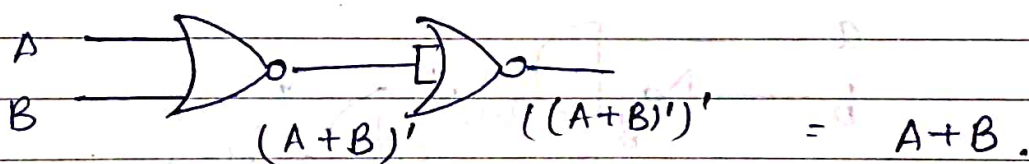
NOT gate using NOR



AND Gate using NOR



OR Gate using NOR



- As well as only by using NOR gate all primary logic gates can be implemented.
- If primary logic gate can be implemented implies any boolean expression can be implemented as LOGIC CIRCUIT.

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Implementing boolean expression using only NAND / NOR logic gate.

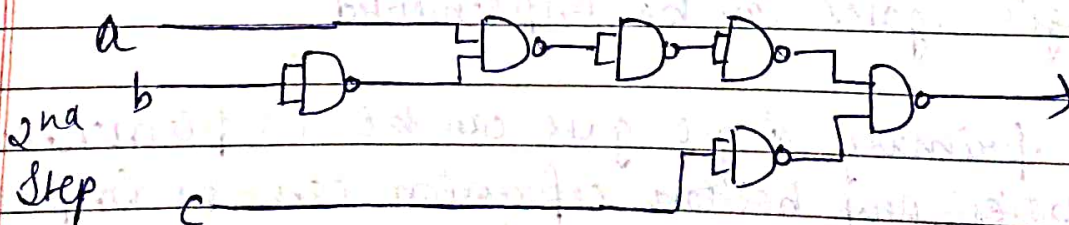
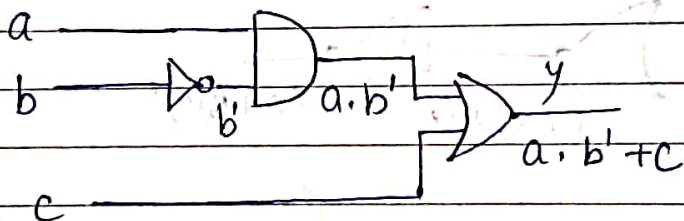
Step 1 - design logic circuit for the given boolean function using primary logic gate.

Step 2 - Replace each and every primary logic gate with its equivalent NAND / NOR logic gates circuit.

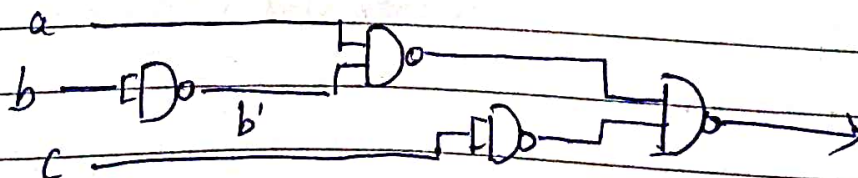
Step 3 - If there are two single input NAND / NOR logic gates connected continuously ignore them.

$$Y = a \cdot b' + c$$

using NAND



3rd



$$a \cdot b' + c$$

Using NOR

